ASP Problem Set 1

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Question 5

(a) and (b)

Suppose x is the number of actual tea-first cups correctly classified by the participant. $50 - x$ is the number of tea-first cups misclassified as tea-first. The experiment process can be conceptualized as selecting x actual tea-first cups that are correctly identified from the 50 actual tea-first cups, and then selecting $50 - x$ actual milk-first cups that are misclassified as tea-first from the 50 actual milk-first cups. Therefore, the probability of the participant classifying k tea-first cups correctly is:

$$P(X = x) = \frac{\binom{50}{x}\binom{50}{50-x}}{\binom{100}{50}}$$

This is essentially the PDF of a hypergeometric distribution. Recall that: In other words:

$$X \sim \text{Hypergeometric}(N = 100, K = 50, n = 50)$$

where N is the total number of cups, K is the total number of actual tea-first cups, and n is the total number of cups classified as tea-first.

```r
library(parallel)

# Settings
B <- 100000
S_obs <- 67

# Set up parallel cluster
n_cores <- detectCores() # 8 cores in total
cl <- makeCluster(n_cores)

# Reproducible parallel RNG
clusterSetRNGStream(cl, iseed = 4210)

# Split simulations across cores
B_each <- rep(B %/% n_cores, n_cores)

# Monte Carlo simulation
sim_list <- parLapply(
  cl,
  B_each,
  function(nsim) {
```

```r
# Number of correctly identified tea-first cups X
# As shown above, X follows a hypergeometric distribution

    X <- rhyper(
      nsim,
      m = 50,
      n = 50,
      k = 50
    )

    # Total number correctly classified (both tea-first and milk-first)
    2 * X
  }
)

stopCluster(cl)

S_sim <- unlist(sim_list)

head(S_sim)
```

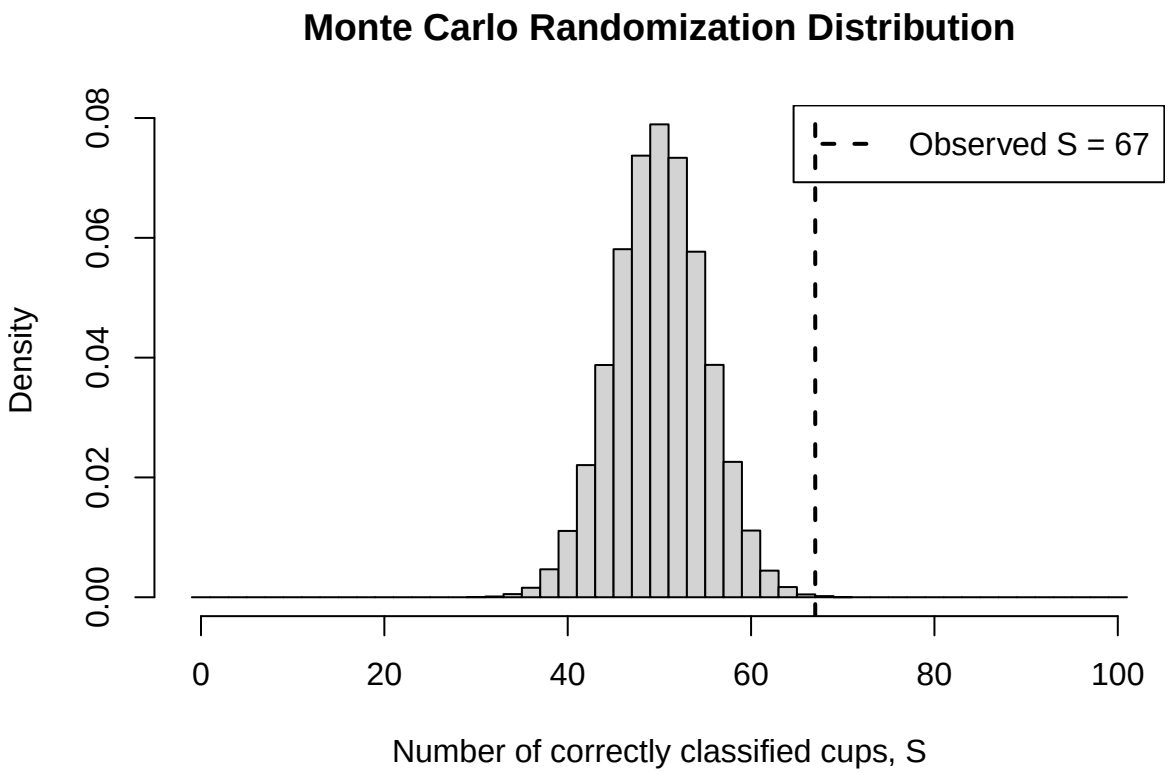
```
## [1] 50 50 54 54 48 48
```

(c) Plot the randomization distribution

```r
# Plot randomization distribution
hist(
  S_sim,
  breaks = seq(-1, 101, by = 2),
  probability = TRUE,
  main = "Monte Carlo Randomization Distribution",
  xlab = "Number of correctly classified cups, S"
)

abline(
  v = S_obs,
  lwd = 2,
  lty = 2
)

legend(
  "topright",
  legend = paste("Observed S =", S_obs),
  lty = 2,
  lwd = 2
)
```



(d) Estimate the one-sided p-value

```
# Monte Carlo p-value
p_hat <- mean(S_sim >= S_obs)

cat("The p-value from simluation is", p_hat,"\n")
```

```
## The p-value from simluation is 0.00041
```

```
if (p_hat < 0.01) {
  cat("We should reject the null hypothesis at 1% significance level.")
} else{
  cat("We cannot reject the null hypothesis at 1% significance level.")
}
```

```
## We should reject the null hypothesis at 1% significance level.
```

(e) Assessment

Since the p-value is even less than 0.01, I would reject the null hypothesis. There is very strong statistical evidence that the participant's classifications depend on whether tea or milk is poured first.